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Type : ACS - 3

Q1) (A) Create a class student with attributes names and marks. Create 3 objects (instance). Write a method display() to print student details. call the method using each object.

code :

class student:

def __init__(self, name, mark):

self.name = name

self.mark = mark

def display():

print("Name :", self.name)

print("Mark :", self.mark)

print("-----")

s1 = student("Arun", 85)

s2 = student("priya", 90)

s3 = student("Kiran", 78)

s1.display()

s2.display()

s3.display()

Output :

name : Arun

mark : 85

name : priya

mark : 90

name : Kiran

mark : 78

- Q1. Create a class calculator that has methods like
(B) add(a,b), subtract(a,b), multiply(a,b) and
division(a,b). create object and call all methods.

code:

class calculator:

def add(self, a, b):
return a+b

def subtract(self, a, b):
return a-b

def multiply(self, a, b):
return a*b

def division(self, a, b):
if b!=0:
return a/b

else:
return "Divided by zero is not allowed"

calc = calculator()

print("Addition :", calc.add(10, 5))

print("Subtraction:", calc.subtract(10, 5))

print("Multiplication", calc.multiply(10, 5))

print("Division :", calc.division(10, 5))

output:

Addition: 15

Subtraction: 5

Multiplication : 50.

Division : 2.0.

Q3) Create a class car. Use a constructor (-init-) to initialize brand, model and price. Display details using a method.

code:

class Car:

def __init__(self, brand, model, price):

self.brand = brand

self.model = model

self.price = price

def display(self):

print("Brand:", self.brand)

print("Model:", self.model)

print("Price:", self.price)

c = Car("Toyota", "Corolla", 1200000)

c.display()

Output:

Brand : Toyota

Model : Corolla

Price : 1200000

Q4) Create a class department. It contains multiple employee objects. Add method to add employee and display all employees. Show delegation by calling employee methods through department class.

code:

```
class Employee:
```

```
    def __init__(self, name, emp_id, salary):
```

```
        self.name = name
```

```
        self.emp_id = emp_id
```

```
        self.salary = salary
```

```
    def display(self):
```

```
        print("Employee id:", self.emp_id)
```

```
        print("name:", self.name)
```

```
        print("salary:", self.salary)
```

```
class Department:
```

```
    def __init__(self, dept_name):
```

```
        self.dept_name = dept_name
```

```
        self.employees = []
```

```
    def add_employee(self, emp):
```

```
        self.employees.append(emp)
```

```
    def display_employees(self):
```

```
        print("Department:", self.dept_name)
```

```
        print("Employee List")
```

```
        for emp in self.employees:
```

```
            emp.display()
```

```
dept = Department("IT")
```

```
e1 = Employee("Anu", 101, 50000)
```

```
e2 = Employee("Priya", 102, 60000)
```

```
e3 = Employee("Kiran", 103, 55000)
```

```
dept.add_employee(e1)
```

```
dept.add_employee(e2)
```

```
dept.add_employee(e3)
```

dept. display_employee()

output :

Department : IT

Employee List

Employee ID : 101

Name : Arun

Salary : 50000

Employee ID : 102

Name : priya

Salary : 60000

Employee ID : 103

Name : Kiran

Salary : 55000

Q5) Create a simple GUI using tkinter. Input fields for two numbers. Add Buttons to Add, subtract, Divide and Multiply. Display result in label. use class based approach.

code :

```
import tkinter as tk
```

```
class calculatorGUI:
```

```
    def __init__(self, root):
```

```
        self.root = root
```

```
        self.root.title("Simple Calculator")
```

```
        self.num1_label = tk.Label(root, text="Number1")
```

```
        self.num1_label.grid(row=0, column=0)
```

```
        self.num1_entry = tk.Entry(root)
```

```
        self.num1_entry.grid(row=0, column=1)
```


self.num2_label = tk.Label(root, text="number 2")

self.num2_label.grid(row=1, column=0)

self.num2_entry = tk.Entry(root)

self.num2_entry.grid(row=1, column=1)

self.result_label = tk.Label(root, text="Result:")

self.result_label.grid(row=2, column=0, columnspan=2)

tk.Button(root, text="Add", command=self.add).grid(row=3, column=0)

tk.Button(root, text="Subtract", command=self.subtract).grid(row=3, column=1)

tk.Button(root, text="Multiply", command=self.multiply).grid(row=4, column=0)

tk.Button(root, text="Division", command=self.divide).grid(row=4, column=1)

def add(self):

a, b = self.get_values()

if a is not None:

self.result_label.config(text="Result: " + str(a+b))

def subtract(self):

a, b = self.get_values()

if a is not None:

self.result_label.config(text="Result: " + str(a-b))

def multiply(self):

a, b = self.get_values()

if a is not None:

self.result_label.config(text="Result: " + str(a*b))

```
def divide(self):
    a, b = self.get_values()
    if a is not None:
        if b != 0:
            self.result_label.config(text = "Result: " + str(a/b))
        else:
            self.result_label.config(text = "cannot divided by zero")
```

```
root = tk.Tk()
app = calculator GUI (root)
root.mainloop()
```

output.

Number 1	10
Number 2	5
Result : 15	
Add	Subtract
Multiply	Division

Q6) Create a To-Do application using tkinter with facility to Add task, Remove task Display all task, store task in list, use class based structure.

code :

```
class TodoApp:
    self.root = root
    def __init__(self, root):
        self.root = root
        self.root.title("To Do App")
```



```
self.tasks = []
```

```
self.task_entry = tk.Entry(root, width=30)
```

```
self.task_entry.grid(row=0, column=0,  
                      columnspan=2)
```

```
tk.Button(root, text="Add task", command=  
          self.add_task).grid(row=1, column=0)
```

```
tk.Button(root, text="Remove Task", command=  
          self.remove_task).grid(row=1, column=1)
```

```
self.task_listbox = tk.Listbox(root, width=40)
```

```
self.task_listbox.grid(row=2, column=0,  
                       columnspan=2)
```

```
def add_task(self):
```

```
    task = self.task_entry.get()
```

```
    if task:
```

```
        self.tasks.append(task)
```

```
        self.update_listbox()
```

```
        self.task_entry.delete(0, tk.END)
```

```
def remove_task(self):
```

```
    try:
```

```
        index = self.task_listbox.curselection()[0]
```

```
        self.tasks.pop(index)
```

```
        self.update_listbox()
```

```
    except:
```

```
        pass
```

```
def update_listbox(self):
```

```
    self.task_listbox.delete(0, tk.END)
```

```
    for task in self.tasks:
```

```
        self.task_listbox.insert(tk.END, task)
```



```
root = Tk()
app = TodoApp(root)
root.mainloop()
```

output :

test.	
add task .	remove task
test1	
test2	
test3	
test4	